



Spatiotemporal LULC change detection and future prediction for the Mand catchment using MOLUSCE tool

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Abstract

The changes in land use and land cover (LULC) are one of the primary forces for the worldwide climate, ecosystem, and environmental changes. A detail understanding of the dynamics of LULC changes is requisite for natural resource management and sustainable development. The present study region is the Mand catchment situated in Chhattisgarh, which has witnessed changes in LULC due to a rise in anthropogenic activities, such as an increase in population, urbanization, mining activity, and agriculture, as well as natural causes, such as climate change. The present study focused on spatiotemporal LULC change analysis and the subsequent prediction of future LULC patterns. A supervised image classification method in Geographic Information System (GIS) was used to analyze decadal LULC changes from 2001 to 2021. The Cellular Automata–Artificial Neural Network (CA–ANN) model, incorporated in the MOLUSCE (Modules of Land Use Change Evaluation) plugin of QGIS, was used for the future prediction of LULC from 2030 to 2040. The overall accuracy of LULC for 2001, 2010, and 2021 was obtained as 82, 86, and 90%, respectively, and the overall kappa coefficient was obtained as 0.79, 0.84, and 0.88, respectively. The decadal study of LULC change concludes that the agricultural land counts highest area of 29.21% as compared to other LULCs in 2001 which was further increased 31.76% in 2021. Increasing trend were also noticed for the open forest, shallow waterbody, fallow land, and settlement for the decadal years 2001 to 2021 by 1.7, 7.41, 7.57, and 2.55%, respectively. A decreasing trend was observed in LULC changes during the decadal years 2001 to 2021 for dense forest, deep water body, and barren land by 10.28, 0.66, and 10.24%, respectively. The LULC predictions for 2030 and 2040 indicate a similar trend to the prior years, with an increase in settlement, fallow land, agricultural land, open forests, shallow waterbodies, scrubland, and a decrease in dense forests, deep waterbodies, and barren land. This comprehensive analysis of the changes in LULC over an extended period will prove to be a valuable resource for policymakers and planners seeking to achieve sustainable development and effective management of the ecosystem within the study area.

Keywords LULC · Remote sensing and GIS · CA–ANN · Accuracy assessment

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Introduction

Land use encompasses the range of human activities involved in the purposeful development and exploitation of land resources. It encompasses various categories, including industrial, agricultural, residential, and transportation land, among others. Conversely, the idea of land cover refers to the physical extent of the Earth's surface, which includes not only its natural features but also the structures built by humans (Sun et al 2023). Land use involves the diverse range of materials that cover the Earth's surface, including crops, forests, grasslands, and buildings. It is a dynamic process occurring on the planet's surface, whereas land cover represents the outcomes of various surface processes. Whether on

a local, national, or global scale, alterations in land use consistently drive accelerated variation in land cover (Sun et al. 2023; Tumer et al. 1994). Land use and land cover (LULC) cannot be distinguished from one other, and it is assessed together (Kafy et al. 2020; Celik et al. 2019; Das et al. 2021).

Change in LULC has become an essential field of study in environmental science, geography, and urban planning due to its significant effects on ecosystems, climatic patterns, and human well-being (Roy et al. 2022). The phenomenon is based on the evolution and changes in Earth's surface, such as changes to vegetation cover, land management practices, and expansion of urbanization. The Earth's land surface has seen significant changes because of rapid urbanization, growth in population, and economic development occurring around the world (Olorunfemi et al. 2020). These modifications have wide-ranging effects on biodiversity, water supplies, soil quality, and the overall health of ecosystems. For promoting sustainable land management practices, reducing environmental risks, and fostering harmonious human–environment interactions, it is essential to comprehend the causes, processes, and consequences of LULC (Kyriakopoulos et al. 2023). The projection by United Nations is that the population of the world will increase by around 60% by 2050 (Liu et al. 2019; Chauhan and Gordon 2020). To date, urban population expansion has been observed in almost every nation in the world, and it has been cited as one of the main causes of abrupt changes in LULC (Rawat and Kumar 2015). Currently, there is a transition of multiple natural land covers into urban development for residential and other anthropogenic applications (Liu et al. 2019; Rawat and Kumar 2015). Worldwide losses of forests and grasslands during the past three centuries have been 19% and 8%, respectively, although human buildups have only occupied 6% of the Earth's surface (Anand and Oinam 2020).

The unexpected alterations in LULC have a damaging impact on the natural environment which results in raising surface temperature, relative humidity, rate of evaporation, and unpredictable rainfall, which influence biodiversity, water quality, quality of soil, and ecosystems (Kafy et al. 2020). Such changes promptly degrade the landscape and have an impact on the land's surface, which increases greenhouse gas emissions, and biodiversity loss, degrades soil resources, and contributes to global climate change (Rahel et al. 2018; Hamad et al. 2018; Majumder et al. 2023; Liu et al. 2019). To replicate future LULC change in this situation, it is essential to determine the causes of these changes (Ruben et al. 2020). For the most effective urban planning and efficient management of natural resources, a thorough investigation of LULC change detection is required, which can be done by utilizing RS techniques on satellite imagery (Mallupattu et al. 2013). LULC modeling is crucial to understanding the LULC process and its variations in the future (Ruben et al. 2020).

To formulate suitable plans for land use management, it is imperative to ascertain the potential future LULC. Predicting future changes through LULC modeling approaches involves considering past LULC transformation patterns, population expansion, topographical features of the area, elevation patterns, and various other factors. The simulation models adopt a probabilistic technique to project future changes (Sarkar and Chouhan 2019; Kafy et al. 2020; Rahman et al. 2017). Models based on mathematical equations, hybrid models, statistical models, Markov models, cellular automata (CA) models, Modules for Land Use Change Simulations (MOLUSCE), and multiagent models are among the various types of models employed for analyzing and forecasting the LULC. These models focus on understanding the causes and effects of LULC change, allowing researchers to examine the underlying dynamics and patterns associated with land use changes (Yang et al. 2012; Subedi et al. 2013; Singh et al. 2015).

The MOLUSCE plugin is the most effective used model for projecting the future pattern of LULC using data from prior years (Kamaraj and Rangarajan 2022; Amgoth et al. 2023; Muhammad, et al. 2022; Mas et al. 2014). A recently released QGIS-based tool called the MOLUSCE-plugin is intended for the investigation of potential LULC scenarios. This plugin offers several functionalities, including the generation of a transition probability matrix using the Cellular Automata–Artificial Neural Network (CA–ANN) approach, as well as simulation-based modeling using various techniques, such as logistic regression (LR), multi-criteria evaluation (MCE), artificial neural network (ANN) and weights of evidence (WoE), at the last stage of simulation, cellular automata (CA) model is employed to estimate and predict the future LULC patterns (Jogun 2016).

The study region is a peri-urban watershed that is experiencing rapid population increase as well as agricultural, industrial, and mining activity. These parameters have helped LULC patterns to continuously alter throughout time. The population of the Mand catchment has increased by 18.02% during the last ten years, showing the area's rapid growth. This development can be linked to the region's economic and political significance, which has sparked the rise of the area (Chandramouli et al. 2011). The primary cause of the elevated level of urbanization within the Mand catchment region can be attributed to the phenomenon of rural-to-urban migration. This is due to the presence of industries and factories within the study area, which provides a compelling reason for most migrants to settle permanently within the city (Gole and Ekka 2019; Wang et al. 2022a, b). This trend is further compounded by the region's excellent transportation infrastructure, convenience, and good standard of living, which collectively contribute to an increase in the total population of the region. Consequently, there has been a swift transformation of the once barren land and forest areas within the catchment into residential zones, mining

sites, and various urban settings. This rapid change in land use has reshaped the landscape, reflecting the evolving needs and activities within the region (Pholsena 2018; Sangha et al. 2019). The improper management of available land has raised substantial concerns regarding the future of LULC. This mismanagement could have detrimental effects on the quality of life within the area. The consequences may include increased risks of natural disasters such as landslides and floods due to unregulated settlements, industries, and factories with inadequate sewage systems, as well as the potential for fires caused by congested streets, among other challenges. Therefore, it became paramount to clearly understand the historical and present scenario of LULC changes to predict the future LULC change which is an effective method to develop a sustainable land use management plan that can help in the years to come to minimize human-induced environmental consequences (Faisal et al. 2018).

This research was carried out within the Mand catchment of the Mahanadi basin, focusing on the examination of geographical factors influencing distinctive LULC change processes. The study explores the conversion of these geographical drivers into an accurate simulation and prediction model for future land changes, employing the CA-ANN model, a novel approach not previously utilized in the area under investigation. The primary objectives of this study are threefold: (1) to analyze the spatiotemporal changes in LULC within the Mand catchment between 2001 and 2021, (2) to employ the CA-ANN model for forecasting LULC in the years 2030 and 2040, and (3) to demonstrate the potential impact of these changes on future land-use planning. It is noteworthy that, to date, no prior research in the region has employed the CA-ANN model for predicting future LULC changes. It is crucial for regulators, resource managers, and government bodies to actively monitor these spatiotemporal models to anticipate shifts in landscape structure and ensure the sustainable utilization of natural resources.

Materials and methods

Study area

The present study region is situated in Chhattisgarh, India, precisely within the Mand catchment of the Mahanadi basin. The geographical coordinates of the study area range from 82°50'54.5" E to 83°36'1.3" E in longitude and 21°42'15.53" N to 23°4'19.75" N in latitude (Fig. 1). The geographical area is comprising of 4730 km², where the elevation is ranges lies between 208 to 1147 m above MSL. The Mand River originates from Surguja district and flows over Jashpur, Korba and Raigarh districts and meets at the Mahanadi River. It is rapidly growing as an industrial area of Chhattisgarh. It is one of the major producers of steel and power in

the country. Agricultural land, dense forests, and arid territory make up the majority of the LULC along the river area. In the agricultural sector, the main crops grown in the Kharif season are paddy, arhar, and maize, whereas wheat, linseed, and chickpea are grown during the Rabi season (Baghel et al. 2023a).

A subtropical monsoon climate, with the monsoon, summer, and winter being three distinct seasons, prevails in the studied area. The winter season lasts from October to February, the summer season is in full force from mid-March through mid-June, and the southwest monsoon starts in June and lasts until mid-September. The southwest monsoon is the primary source of groundwater recharge in the study area, producing most of the annual precipitation (85%) with an average precipitation total of 1192.1 mm. The lowest temperature was 8.2 °C in January, and the average maximum temperature was 42.5 °C in May (Baghel et al. 2021).

Data acquisition

The satellite images were acquired by considering the phenological effects, seasonality, and cloud cover. In this research, satellite images from Landsat Thematic Mapper TM 5 were collected for 2001 and 2010 and Landsat 8-OLI for 2021 with a spatial resolution of 30×30 m (<https://earthexplorer.usgs.gov/>).

The Shuttle Radar Topography Mission (SRTM)'s Digital Elevation Model (DEM), with a spatial resolution of 30 m, was obtained from the United States Geological Survey (USGS) (<http://www.earthexplorer.usgs.gov>). The DEM was utilized to demarcate the catchment and slope using spatial analytical tools within the ArcGIS software. The road data system's criticality highlights the significance of its derivation from Open Street Map (<http://www.openstreetmap.org/>) and used for future LULC prediction by CA-ANN model. Details of all input parameters are discussed in (Table 1).

The acquired data set was pre-processing in ArcGIS 10.5 software by techniques, namely, (a) re-projection of the images from World Geodetic System 1984 (WGS84) to Universal Traverse Mercator (UTM) Zone 44 N, (b) layer stack and mosaic of the image, (c) extraction of images for the study region, and (d) supervised image classification employing maximum likelihood method and the LULC changes. The future prediction was done using CA-ANN present in the MOLUSCE plugin in QGIS.

To achieve accuracy, the modeling parameters must be precisely selected. Several studies (Alam et al. 2021; Hakim et al. 2019; Saputra and Lee 2019) have examined different variables, such as road distance, distance from the river, distance from built-up area, population density, DEM, and slope, to investigate LULC patterns. In the present study, the LULC map of 2001, and 2010 were selected as an initial and

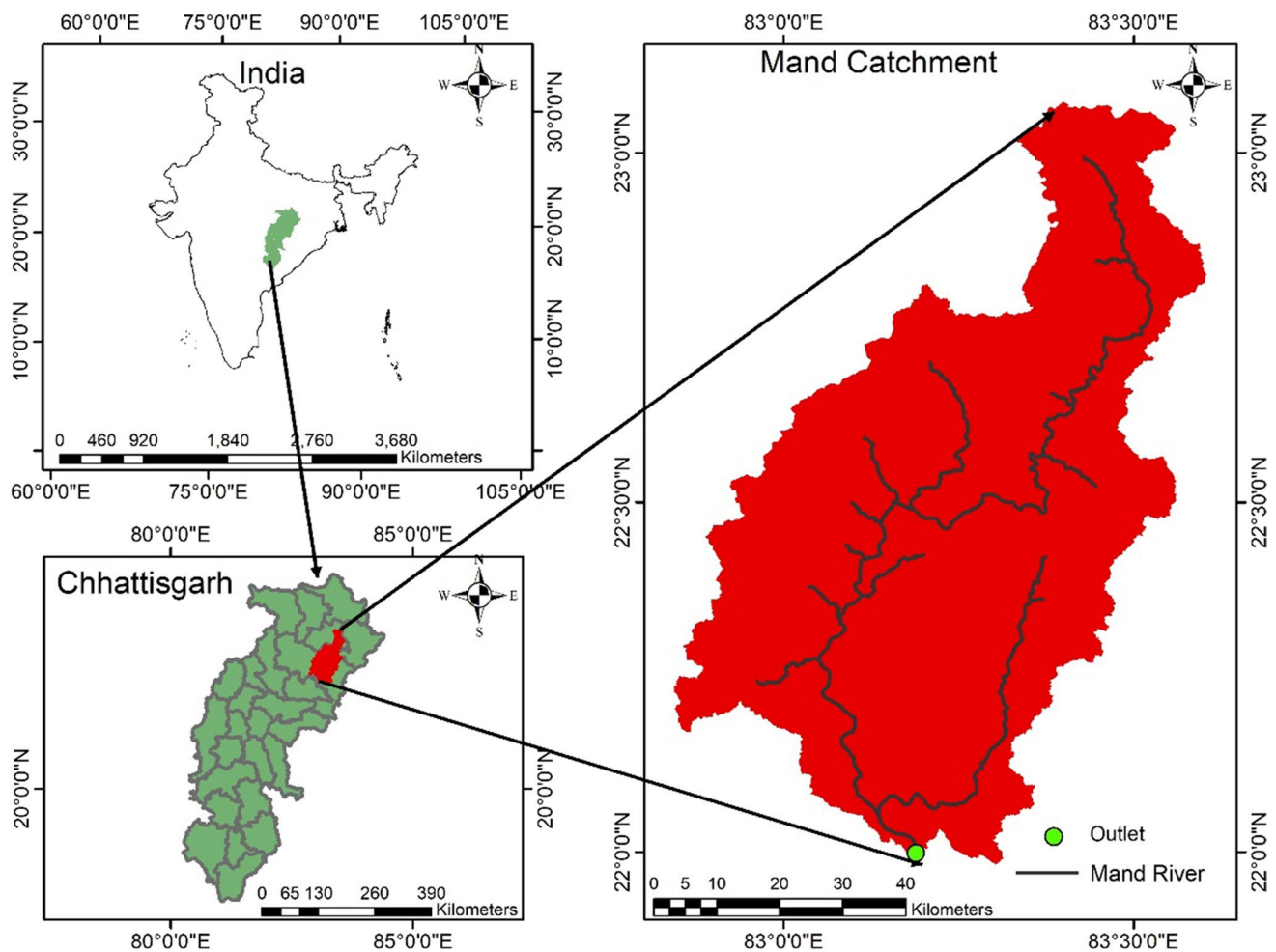


Fig. 1 Location of the study area

Table 1 Various input parameters of the research

Sr. no.	Item	Type	Source
1	Open street map	Road Network (spatial vector data)	http://www.openstreetmap.org/
2	Digital Elevation Model (DEM)	SRTM	http://www.earthexplorer.usgs.gov
3	Satellite image	Landsat Thematic Mapper TM 5 (2001, 2010) and Landsat 8-OLI (2021) (30 m resolution)	http://www.earthexplorer.usgs.gov

final layer, respectively, and the spatial variable maps such as DEM (Fig. 2a), slope (Fig. 2b), and distance from the road (Fig. 2c) were considered to check the geometry correction in MOLUSCE in QGIS.

LULC classification

The categorization was performed through the utilization of a supervised classification technique, specifically the maximum likelihood algorithm, within the ArcGIS

framework. (Rawat and Kumar 2015; Chilagane et al. 2020; Shiferraw et al. 2019; Baghel et al. 2023b). On average 300–350 training samples were selected for each LULC class, these training samples have been selected based on the user knowledge and Google Earth time series images. This study classified LULCs into nine classes, Agricultural land (AL), Barren land (BL), Fallow land (FL), Dense Forest (DF), Open Forest (OF), Scrubland (SL), Shallow water body (SWB), Deepwater body (DWB), and Settlement (SM), as shown in Table 2.

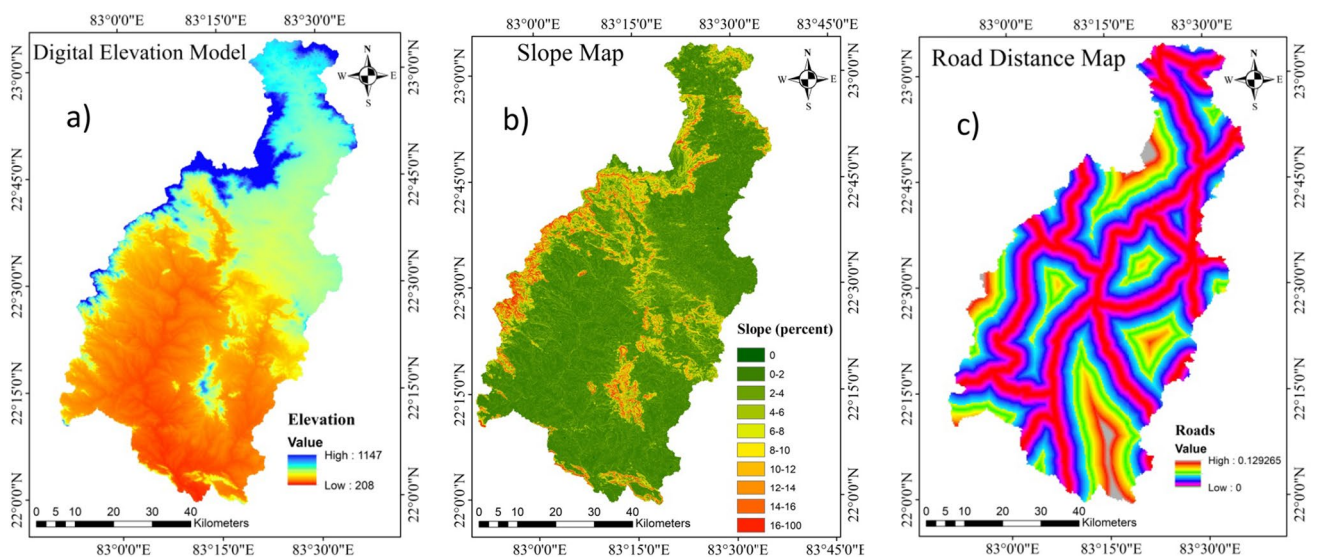


Fig. 2 a–c Spatial variable maps

Table 2 Definition of LULC classification types

LULC classes	Description
Agricultural land (AL)	Land utilized for agricultural purposes encompasses the cultivation of crops, such as paddy and wheat, amongst others
Barren land (BL)	Sand, rock, and other empty lands
Fallow land (FL)	Temporary cultivable land, which may potentially remain fallow for one or more growing seasons
Dense Forest (DF)	Untouched forests, unscathed by human activities, exhibit high biodiversity and boast a vegetation cover of over 70% in the area. Furthermore, the forests contain trees that exceed the height of 10 m. (Watson et al. 2018)
Open Forest (OF)	A forest that has undergone prolonged destruction from human interventions has led to a significant depletion of biodiversity, loss of various species, and a significant reduction in vegetation coverage, estimated to be within the range of 10–15%. (Faichia et al. 2020)
Scrubland (SL)	All forested areas exhibit poor tree development, characterized by a prevalence of diminutive or stunted trees with a canopy density measuring less than 10%
Shallow water body (SWB)	The water is of sufficient depth, such that the topography of the underlying surface has a discernible influence on the propagation of surface waves
Deepwater body (DWB)	Surface water, whether contained within impoundments, such as ponds, lakes, or reservoirs, or in motion as streams, rivers, and other aqueous entities
Settlement (SM)	The available land for construction encompasses various sectors, such as marketable services, industry, warehouses, residences, public administration, public services, and transportation, comprising buildings, houses, factories, and roads

Accuracy assessment

The accuracy assessment of post-classified images was performed to ascertain the count of pixels that were excluded, misclassified, and correctly classified within an image (Congalton and Green 2019). The researchers employed the heuristic stratified random sampling technique to select the reference pixels that were evenly distributed across the LULC

classes within the study area (Chen et al. 2018). These pixels were independent of the sites chosen during the training phase. For each year under consideration, a total of 720 reference pixels (about 80 per class) including all LULC classes were selected for accuracy assessment. The different accuracy indices such as user’s accuracy, producer accuracy, overall accuracy, and Kappa coefficients (K) were estimated using Eqs. (1), (2), (3), and (4), respectively:

$$\text{User's accuracy} = \frac{\text{Number of corrected classified pixels in each category}}{\text{total number of reference pixels in that category (the row total)}} \times 100$$

(1)

$$\text{Producer accuracy} = \frac{\text{Number of correctly classified pixels in each category}}{\text{total number of reference pixels in that category (column total)}} \times 100 \quad (2)$$

$$\text{Overall accuracy} = \frac{\text{Number of corrected classified pixels (diagonal)}}{\text{Total number of reference pixel}} \times 100 \quad (3)$$

$$K = \frac{\sum_{i=1}^r x_{ii} - \sum_{i=1}^r (x_{i+} \times x_{+i})}{N^2 - \sum_{i=1}^r (x_{i+} \times x_{+i})} \quad (4)$$

where r = number of rows, N = total number of sites, K = kappa coefficient, x_{ii} = number in rows i and columns i , x_{+i} = total for row i , and x_{i+} = total for the column. The rank criteria of K are presented in Table 3 (Ishak and Bakar 2012).

After ensuring the accuracy of the classified images, the LULC maps were developed for the Mand catchment. The LULC change analysis for 2001 to 2010, 2010 to 2021, and 2001 to 2021 was done in the present study. The methodological flowchart of the research is illustrated in Fig. 3.

Modelling process of future LULC scenarios

The prediction of LULC for the years 2030 and 2040 was conducted utilizing a CA model. CA models are dynamic models that operate from the bottom up, incorporating both the spatial and temporal dimensions, thereby enhancing the modeling process. These models simulate the complex interplay of time and space while considering the discrete nature of space, time, and state variables. (Subedi et al. 2013). These components collectively enable the CA model to effectively demonstrate and simulate the intricate dynamics of LULC prediction research (Eq. 5). The CA model utilizes a filter to identify the neighbors of each cell. In this process, the proximity between the central cell and its neighbors plays a crucial role. In the model, the weight factor assigned to the central cell and its neighbor is directly proportional

to the proximity between them. The closer their distance, the greater the weight factor assigned to them (Wang et al. 2020):

$$S_{t+1} = f(S_t, N) \quad (5)$$

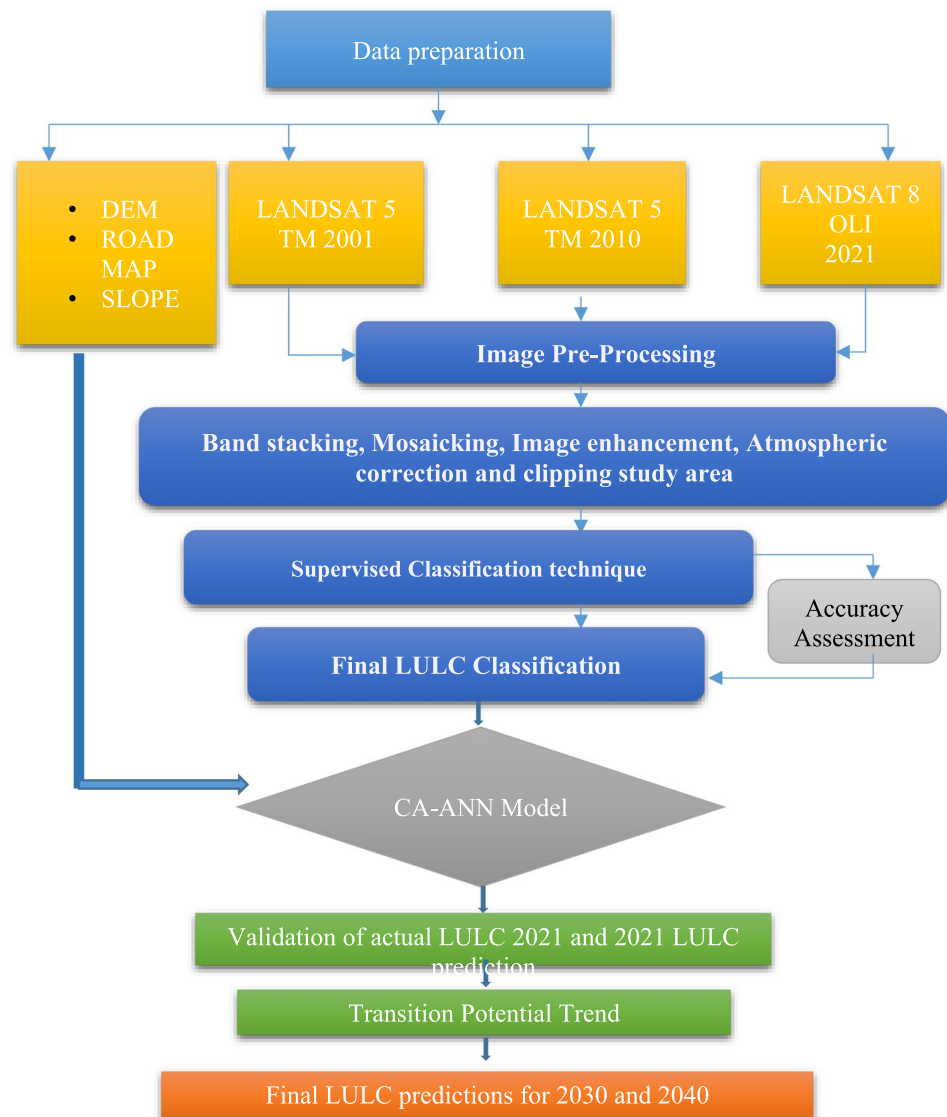
where S is the set of states of the finite cells; t and $t+1$ are the previous years and the next year; N is the neighborhood of cells, and f is the conversion rule of local space.

The transition probabilities from the Multi-Layer Perception–Artificial Neural Network (MLP–ANN) learning procedure (in QGIS 2.18.24 plugin MOLUSCE) are used in the CA model to characterize the LULC changes. A tool named MOLUSCE-plugin, which is based on Quantum Geographic Information System (QGIS), has been released to investigate potential LULC scenarios. The plugin includes functionalities, such as generating a transition probability matrix using the CA–ANN approach and simulation-based modeling with various techniques, such as LR, MCE, WoE and ANN. The last stage of simulation employs a CA model to estimate and predict future LULC pattern (Jogun 2016). The MOLUSCE interface provides an intuitive and user-friendly experience, complete with specialized modules and functions. Starting with the input module, the plugin includes multiple phases, such as a change in area analysis, modeling techniques, prediction, and validation. This particular model presents superior precision in prediction and comprehensively incorporates both the static and dynamic characteristics of LULC alterations. (Olorunfemi et al. 2020).

The distance to the road, slope, and DEM were considered the independent factors, while the historical trend of the LULC changes from 2010 and 2021 was considered the dependent factor. Various techniques, including Cramér's coefficient, joint information uncertainty, and Pearson's correlation, can be used to assess the correlation between geographic parameters in two raster imagery, which is used to investigate the association among spatial components (Hakim et al. 2019). These techniques enable the examination of relationships and associations between the spatial variables obtain by the transition matrix. The transition matrix provides valuable insights into the spatial dynamics and transformations occurring over the specified period. In

Table 3 Rank criteria of K

Sl. no.	K	Statement
1	<0	Poor
2	0.00–0.20	Slight
3	0.21–0.40	Fair
4	0.41–0.60	Moderate
5	0.61–0.80	Substantial
6	0.81–1.00	Almost perfect

Fig. 3 Methodology flowchart of the research

the present study, the CA-ANN model is used to calibrate, simulate, and predict the LULC. At the core of ANN are the interactions between interconnected neurons and the dynamic adjustment of the weight connections linking them. These fundamental components enable the network to process and learn from input data, as the connections' weights are modified based on the network's training algorithm and the desired output. (Bhattacharya et al. 2020). Finally, the following variables, namely, iterations 1000, neighborhood 1, momentum 0.05, hidden layer 12, and learning rate 0.001, were established for forecasting the LULC map for 2021 (Das and Sarkar 2019). The validation of simulated LULC (2021) was done with the actual LULC (2021) map and different coefficients such as correctness coefficient, overall kappa coefficient, historical kappa coefficient, and local kappa coefficient were calculated.

Results and discussion

LULC classification

The categorization was performed through the utilization of a supervised classification technique, specifically the maximum likelihood algorithm, within the ArcGIS framework with the scale of 1:50,000 (NRSC 2019; Rawat and Kumar 2015; Chilagane et al. 2020). On average 300–350 training samples were selected for each LULC class, these training samples have been selected based on the user knowledge and Google Earth time series images. Nine land use/cover classes were identified in the false color composite of the satellite imagery and accuracy assessment was carried out over the classified images to check the statistical stability of the classified images. LULCs into nine classes are

agricultural land (AL), barren land (BL), fallow land (FL), dense forest (DF), open forest (OF), scrubland (SL), shallow water body (SWB), deepwater body (DWB) and settlement (SM). A similar result also reported by the numerous researchers, where they observed that a decrease in dense forest and increase in settlement and agricultural land (Kumar et al. 2021; Gazi and Mondal 2018).

Accuracy assessment of LULC classes

According to the findings of the historical LULC classification 2001, 2010 and 2021 (Fig. 4), it has been observed that both Land-sat TM and Land-sat OLI techniques have randomly chosen an average of 80 points for each LULC

class, to evaluate the accuracy of classification. These selected points/pixels were verified by ground truthing and using Google Earth time series images (Faichia et al. 2020). The overall accuracy of LULC for 2001, 2010, and 2021 was obtained as 82%, 86%, and 90%, respectively, with overall kappa statistics of 0.79, 0.84, and 0.88, respectively (Table 4). A similar study conducted by Uddin et al. (2023) for future LULC predictions using CA-ANN model, reporting an accuracy percentage of 86.49% and a kappa value of 0.83. Other researchers have also reported comparable accuracy results for LULC classification, specifically with similar LULC classes (Wang et al. 2022a, b; Santhosh and Shilpa 2023; Qu et al. 2021; Thakur et al. 2020).

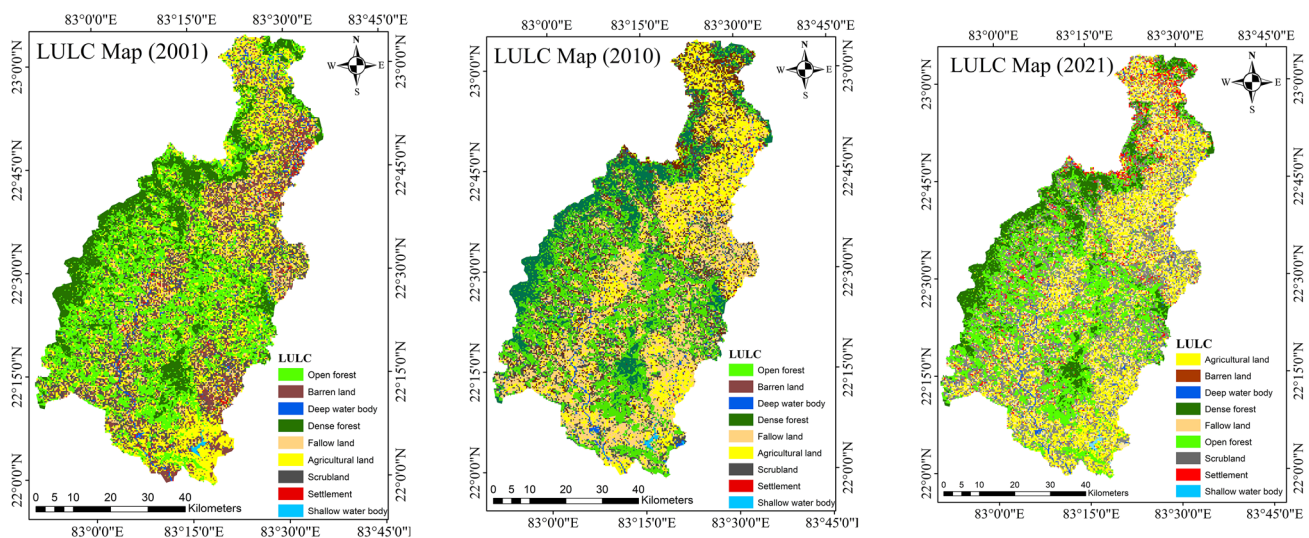


Fig. 4 Land use land cover map of Mand Catchment for the years 2001, 2010, and 2021

Table 4 Results of the LULC classification accuracy for 2001, 2010 and 2021

LULC classes	2021		2010		2001	
	UA (%)	PA (%)	UA (%)	PA (%)	UA (%)	PA (%)
DF	100	100	87.5	100	75	100
SL	80	80	80	100	80	80
OF	100	100	100	71.43	80	57.14
SWB	100	100	80	100	80	100
DWB	100	100	100	83.33	100	83.33
BL	80	80	80	100	80	100
AL	75	85.71	87.5	77.78	75	85.71
FL	80	66.67	60	60	80	57.14
ST	100	100	100	100	100	100
Overall accuracy	90%		86%		82%	
Kappa coefficient	88.68%		84.13%		79.68%	

Analysis of land use/cover change from 2000 to 2021

LULC changes during 2001, 2010, and 2021 were analyzed using supervised classification in the GIS environment. The result of the present research revealed noticeable changes in the LULC between 2001 and 2021. The decadal LULC change analysis shows that the highest decrease in DF, BL, and DWB was 5.62, 5.56, and 0.39%, respectively, from 2001 to 2010, whereas increase in FL, AL, SL, OF, SWB, and ST as 5.55, 2.18, 1.64, 1.03, 0.68, and 0.5%, respectively. Similarly, the maximum changes of LULC from 2010 to 2021 were observed in BL and DF decreases at 0.48 and 4.65%, respectively, whereas a significant increase in agricultural land was observed as 5.78% (Table 5).

In the comparison of LULC from 2001 to 2021, the AL was found to be the largest LULC class in the study area covering 29.21% (1381.61 km²) in 2001 to increased 31.76% in 2021. This was followed by the OF as 841.2 km² (17.78%), DF 789.29 km² (16.69%), BL 592.72 km² (12.53%), and SL 555.11 km² (11.74%) in 2001. Other LULC classes exhibited for smaller proportions of the study region in 2001 are FL covered 425.05 km² (8.97%), SWB covered 65.34 km² (1.38%), DWB covered 42.78 km² (0.90) and ST covered 37.36 km² (0.79%) (Table 6).

The study also reveals a decreasing trend in DF, DWB and BL and an increasing trend for OF, SL, SWB, AL, FL, and ST between the years 2001, 2010, and 2021. Compared to 2001, the percentage of areas covered in 2021 had decreased for DF, DWB, and BL by 10.28, 0.66, and 10.24%, respectively. A percentage increase was observed for OF, SL, AL, FL, SWB, and ST by 1.7, 7.41, 2.55, 7.57, 0.77, and 1.19%, respectively (Fig. 5). Due to the increase in population and anthropogenic activities, such as mining and the construction of roads and buildings, the DF degraded very fast, losing natural balance, and biodiversity which leads to an increase in OF and SL (<http://cgclimatechange.com/forest-and-biodiversit/>; Chakraborty 2023). Similar trends of decrease in DF and OF were also reported in the previous study (Firozjaei et al. 2021; Chatterjee et al. 1994; Saini et al. 2016). Jhum cultivation can also be a major cause of DF reduction and a rise in AL. AL increased continually from 2001 to 2021 due to a rise in population and government policies. The state government provides subsidies for achieving high production targets which motivates farmers to choose agriculture as their occupation (Sun et al. 2023). The noticeable growth in AL and urban areas may be due to the demographic rise which indirectly affects the forest land (Yadav and Tripathi 2023; Bera and Chatterjee 2019; Beckers et al. 2020). DWB decreases due to high soil erosion

Table 5 Decadal change of LULC in Mand catchment for the period of 2001–2021

LULC classes	2001–2010		2010–2021		2001–2021	
	Area in km ²	% changes	Area in km ²	% changes	Area in km ²	% changes
DF	266.05	– 5.62	220.12	– 4.65	– 486.16	– 24.27
SL	– 77.41	1.64	– 273.22	5.78	350.63	17.50
OF	– 48.55	1.03	– 31.86	0.67	80.41	4.01
SWB	– 32.19	0.68	– 4.3	0.09	36.49	1.82
DWB	18.45	– 0.39	12.78	– 0.27	– 31.23	– 1.56
BL	263.12	– 5.56	221.42	– 4.68	– 484.53	– 24.19
AL	– 103.18	2.18	– 17.34	0.37	120.52	6.02
FL	– 262.29	5.55	– 95.02	2.01	357.30	17.83
ST	– 23.64	0.5	– 32.47	0.69	56.11	2.80

Table 6 Area covered by different LULC classes in Mand catchment

LULC classes	2001		2010		2021	
	Area in km ²	% area covered	Area in km ²	% area covered	Area in km ²	% area covered
DF	789.29	16.69	523.24	11.06	303.13	6.41
SL	555.11	11.74	632.51	13.37	905.74	19.15
OF	841.2	17.78	889.75	18.81	921.61	19.48
SWB	65.34	1.38	97.53	2.06	101.83	2.15
DWB	42.78	0.9	24.33	0.51	11.55	0.24
BL	592.72	12.53	329.61	6.97	108.19	2.29
AL	1381.61	29.21	1484.79	31.39	1502.13	31.76
FL	425.05	8.97	687.33	14.53	782.35	16.54
ST	37.36	0.79	61.001	1.29	93.4698	1.98

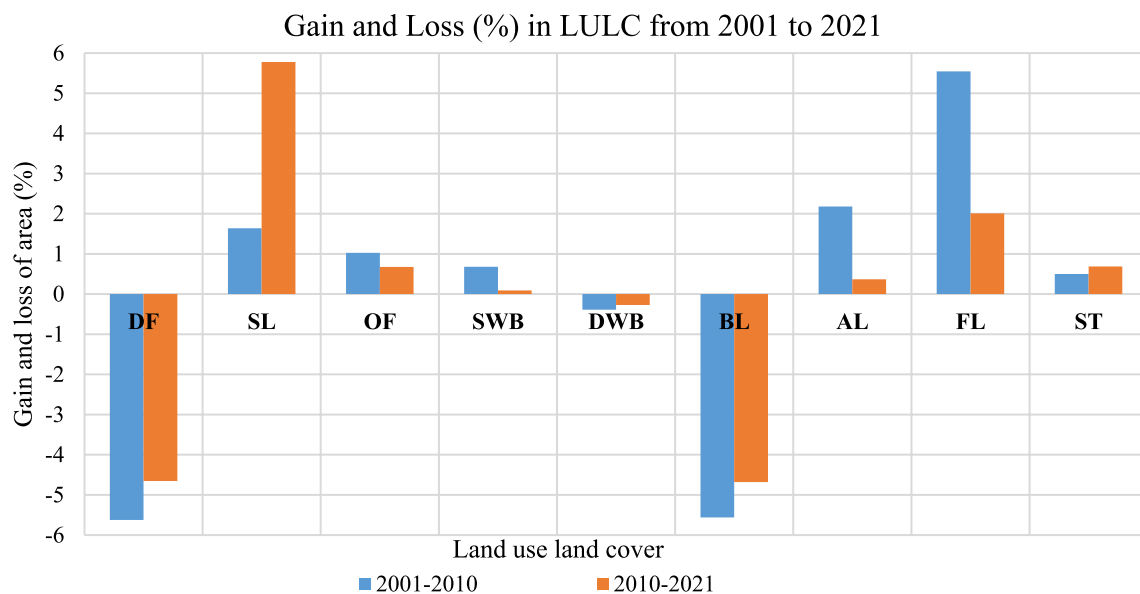


Fig. 5 Area gain and loss for different LULCs from 2001 to 2021

(Wang et al. 2022a, b; NRSC 2019), and the SWB area marginally expanded as a result of the human demand for nearby fishponds and other minor reservoirs for sustaining agriculture. In addition, the increase in SWB and decrease in DWB may be due to the siltation in the river bed (Kasuki et al. 2023; Dash and Maity 2023; Mondal et al. 2018). The decrease in BL was due to a transition from BL to ST and other lands. The expansion in ST was due to the rise in population by an 18.02% increment in the decadal population (Chandramouli et al. 2011).

Future LULC change simulations

Model validation of predicted LULC change in 2021

In the present study, the developed LULC in QGIS was validated for the year 2021. Subsequently, the classification of the given terrain and the variations in LULC between the starting year of 2001 and the ending year of 2010 are quantified and yield a transition matrix that

presents the relative quantity of pixels that shift from one classification to another (Table 7). The simulated and observed LULC for 2021 were compared in the MOLUSCE plugin, and different kappa statistics (% of correctness, K_{overall} , K_{histo} , and K_{loc}) were calculated. During the validation period, % of correctness, K_{overall} , K_{histo} , and K_{loc} was observed as 0.843, 0.867, 0.878, and 0.852, respectively (Table 8). All the Kappa coefficients were above the acceptable criteria of 80%, indicating that the accuracy was adequate for predicting future land use (Regmi et al. 2014; Kamaraj and Rangarajan 2022; Alam et al. 2021; Congalton and Green 2019; Khawaldah 2016). The model was perfectly suitable for predicting LULC for 2030 and 2040, because the findings obtained in terms of both the actual and forecasted LULC for 2021 showed comparable percentages for the rise and decrease in the area (Abijith and Saravanan 2022). A validation graph between the observed 2021 and predicted 2021 LULC map is shown in Fig. 6.

Table 7 Transition matrix for LULC from 2001 to 2010

	DF	OF	SL	DWB	SWB	ST	AL	FL	BL
DF	0.7672	0.0086	0.0000	0.0000	0.0000	0.1897	0.0000	0.0000	0.0345
OF	0.0000	0.6931	0.3003	0.0058	0.0000	0.0007	0.0000	0.0000	0.0002
SL	0.0000	0.0176	0.6062	0.2906	0.0291	0.0023	0.0050	0.0000	0.0492
DWB	0.0000	0.0004	0.0051	0.1451	0.0612	0.0036	0.0232	0.0002	0.7612
SWB	0.0014	0.0007	0.0002	0.0468	0.0116	0.0416	0.8317	0.0055	0.0606
ST	0.0101	0.0000	0.0000	0.0101	0.1924	0.5291	0.2291	0.0101	0.0190
AL	0.0000	0.0000	0.0007	0.0090	0.3012	0.0120	0.3896	0.0013	0.2862
FL	0.0000	0.0000	0.0000	0.0000	0.0102	0.1964	0.5026	0.2474	0.0434
BL	0.0003	0.0114	0.0921	0.2082	0.1576	0.0103	0.2090	0.0001	0.3110

Table 8 Kappa coefficient of LULC predicted (2021)

Indicators	Predicted
% of correctness	0.843
K_{overall}	0.867
K_{histo}	0.878
K_{loc}	0.852

Future LULC simulations for 2030 and 2040

The subsequent LULC projection for the years 2030 and 2040 within the Mand catchment area was acquired by employing the same approach utilized for forecasting LULC in 2021. Figure 7 depicts the predicted LULC of 2030 and

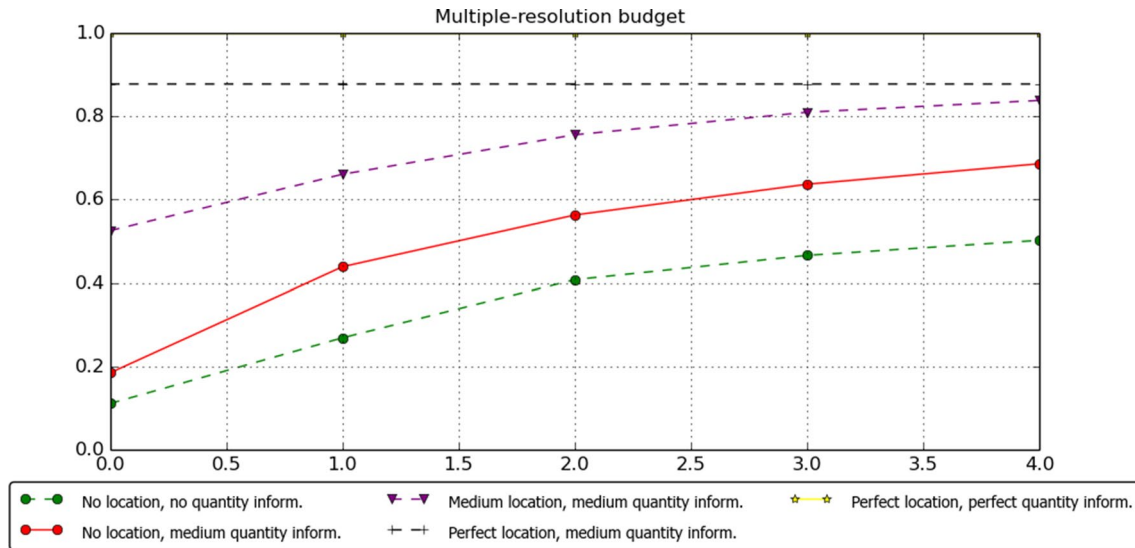
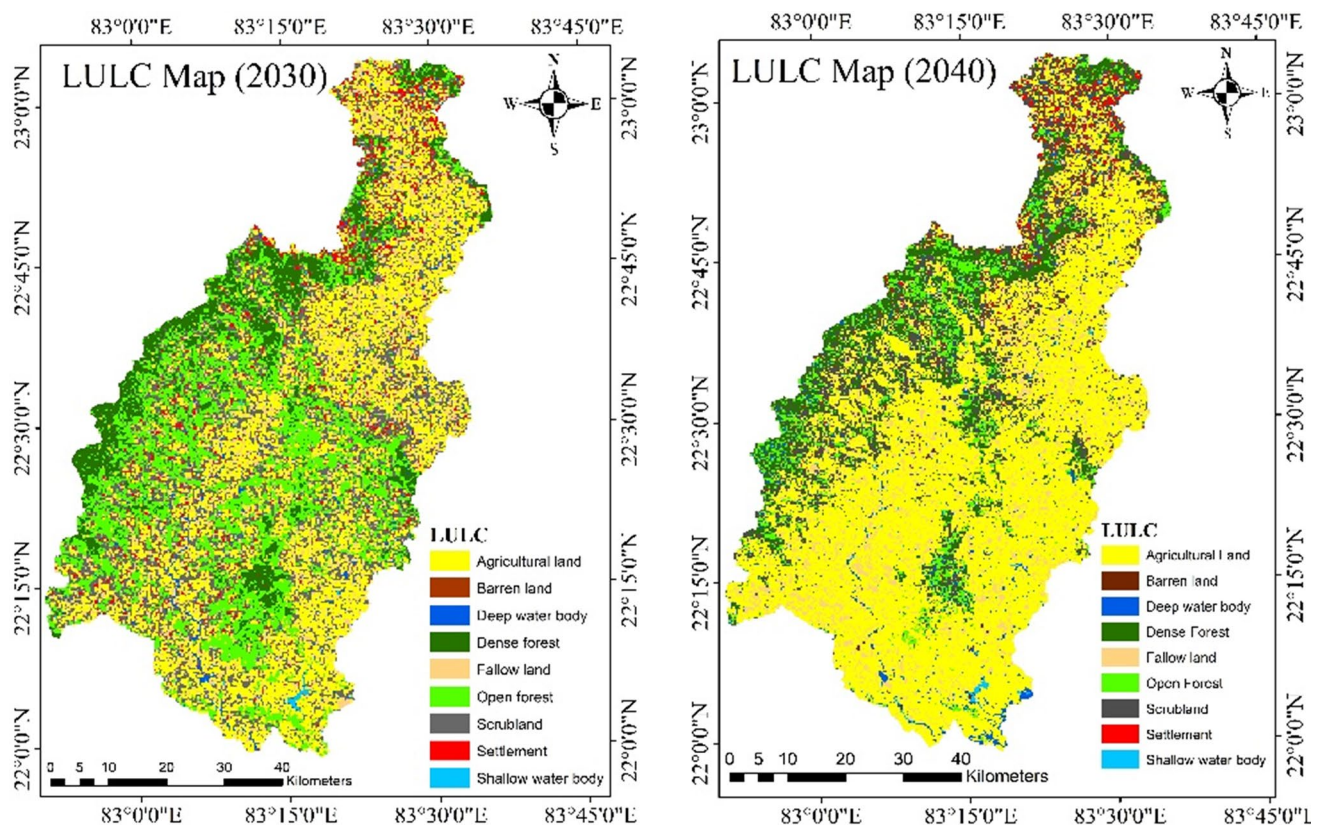
**Fig. 6** Validation graph of observed LULC (2021) and simulated LULC (2021)**Fig. 7** Predicted LULC maps for the years 2030 and 2040

Table 9 Future LULC classification for 2030 and 2040

LULC classes	2030		2040	
	Area in km ²	% area covered	Area in km ²	% area covered
DF	220.21	4.65	172.56	4
SL	946.56	20.01	963.26	20
OF	932.23	19.71	939.95	20
SWB	105.32	2.23	108.36	2
DWB	8.91	0.19	5.04	0
BL	74.95	1.58	48.02	1
AL	1521.98	32.18	1543.31	33
FL	791.54	16.73	802.68	17
ST	128.65	2.72	146.82	3

Table 10 Predicted percent (%) area of LULC changes from 2021 to 2040

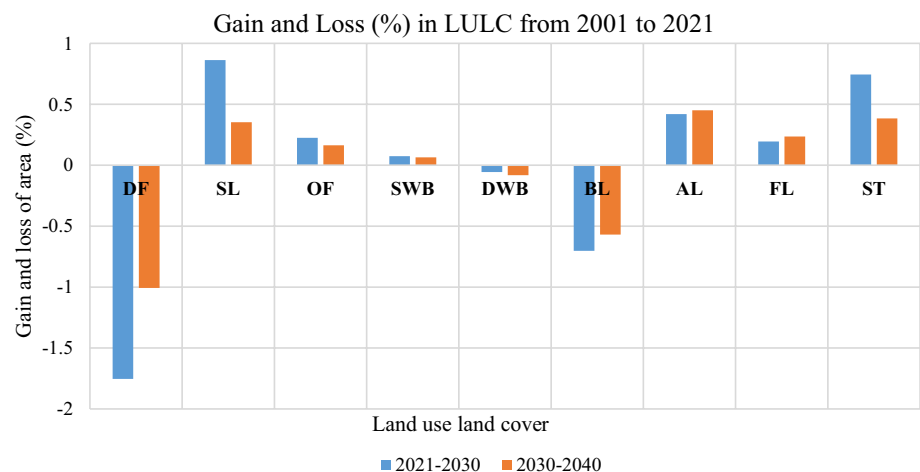
LULC classes	2021–2030		2030–2040	
	Area in km ²	% changes	Area in km ²	% changes
DF	82.92	−1.75	48	−1.01
SL	−40.82	0.86	−17	0.35
OF	−10.62	0.22	−8	0.16
SWB	−3.49	0.07	−3	0.06
DWB	2.64	−0.06	4	−0.08
BL	33.24	−0.7	27	−0.57
AL	−19.85	0.42	−21	0.45
FL	−9.19	0.19	−11	0.24
ST	−35.18	0.74	−18	0.38

2040. The result of the research shows that, drastic changes in dense and open forests, agricultural land, and settlement. From the study, it was found that ST increased in 2030 and 2040, due to a fast-growing population accounting for

128.65 km² (2.72%) and 146.82 km² (3%), respectively (Table 9). However, BL was being reduced over time by becoming developed land. For instance, the simulation predicted that DF would decline by 220.21 km² (4.65%) in 2030 and marginally reduce by 172.56 km² (4%), respectively, in 2040 as a result OF and SL increasing significantly. AL will also slightly increase in 2030 and 2040 by 1521.98 km² (32.18%) and 1543.31 km² (33%) due to an increase in farming activities (Table 10). The final LULC maps prediction for 2030 and 2040 are depicted in Fig. 7, which indicates that DF diminished to a wide extent due to anthropogenic activities and AL increases due to an increase in population which furtherly increases ST. The analysis also considers the changes in LULC as a proportion of the entire land area. A positive figure indicates an expansion in the area of a specific class, whereas a negative value signifies a reduction in the same. (Table 10; Fig. 8). The researchers also noted a similar trend in results, showcasing a successful application of the CA–ANN model for predicting future LULC change patterns (Uddin et al. 2023; Koko et al. 2022; Shahfahad et al. 2022; Mitra et al. 2023).

Conclusions

Assessing changes in LULC and predicting future LULC change analysis is crucial for monitoring of natural resources and urban planning within a watershed. The aim of the present study was to analyze the LULC change and predict the future LULC change pattern for the Mand Catchment. The AL counts highest area of 29.21% in 2001 increased to 31.76% in 2021. OF, SWB, FL, and ST increased by 1.7, 7.41, 7.57, and 2.55%, respectively, from 2001 to 2021. DF, DWB, and BL decreased by 10.28, 0.66, and 10.24%, respectively, over the same period. The LULC maps for 2021 were most accurate with a Kappa coefficient of 0.87. The overall accuracy of LULC for 2001, 2010, and 2021 was obtained

Fig. 8 Predicted loss and gain in the area (%) of different LULC classes for the year 2030, and 2040

as 82, 86, and 90%, respectively, and the overall kappa coefficient was obtained as 0.79, 0.84, and 0.88, respectively.

The LULC predictions for 2030 and 2040 reveal a continuation of trends observed in previous years, with an upward trajectory in ST, FL, AL, OF, SWB, and SL, accompanied by a decline in DF, DWB, and BL. The forecasted LULC map for the year 2040 relative to 2021 showed a decrease in DF and BL by 3 and 1%, respectively, whereas SL, AL, and ST, all are expected to increase by approximately 1% in 2040. The results showed the DF would be converted to OF, SL, FL, and AL due to the increase in anthropogenic activity. A possible reason for decreasing DF is due to demographic influences on the land to be transformed for various uses, such as settlement, agricultural land, mines, or industrial, farmland. This long-term LULC changes study will be helpful to policymakers, and planners for sustainable management of the ecosystem of the study area.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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